MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 15: Laurent Series

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Lecture outline

1. Laurent Series Theorem
2. Proof of Laurent Series Theorem

Laurent Expansion

A function has a Taylor expansion about a point a when it is differentiable in some open set having a as an interior point. We now consider a different kind of series expansion that can be made about a point at which the function is not differentiable, or perhaps not even defined.

The set of points $r < |z - z_0| < R$ is called an *annulus.* r is the radius of the smaller circle, and R is the radius of the exterior circle. We may have $r = 0$ and in this case z_0 is not in the annulus and it is called a punctured disk. We may also have $R = \infty$, that is all points exterior to the circle of radius r about z_0.

Laurent Expansion Theorem. Part 1

Theorem

Let $0 \leq r < R \leq \infty$ and suppose f is differentiable the annulus $r < |z - z_0| < R$. Then for each z in this annulus,

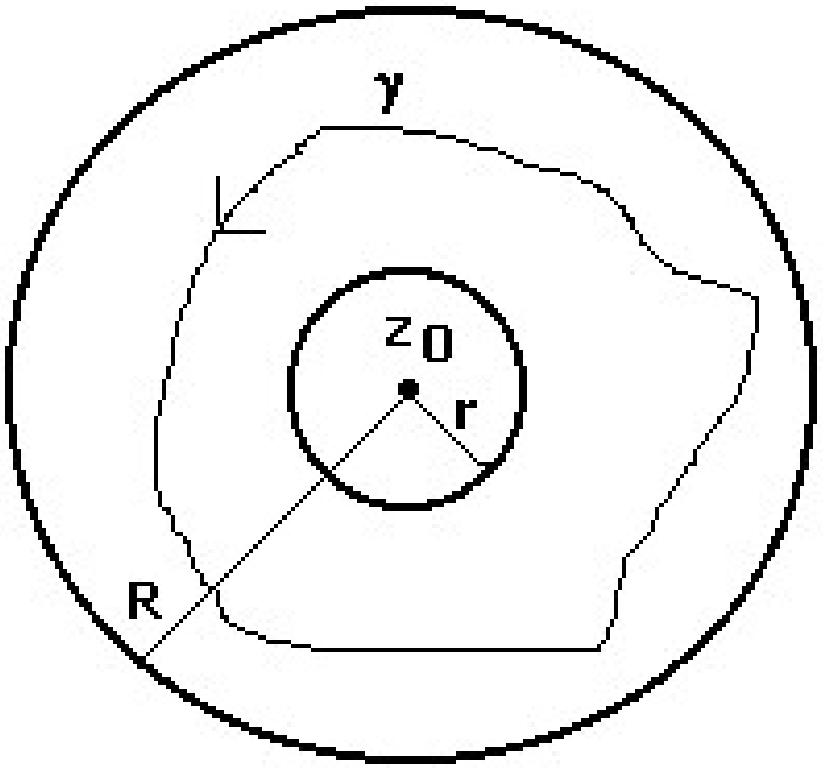
$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \tag{1}$$

where for each integer n

$$c_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz \tag{2}$$

where γ is any closed path about z_0 lying entirely in the annulus.

Laurent Expansion Theorem. Part 2



Laurent Expansion Theorem. Part 3

Remark: The series (1) is called a *Laurent expansion* of $f(z)$ about z_0. Note that written out (1) is

$$\cdots + \frac{c_{-2}}{(z-z_0)^2} + \frac{c_{-1}}{(z-z_0)} + c_0 + c_1(z-z_0) + c_2(z-z)^2 + \cdots$$

$f(z)$ need not be differentiable, or even defined at z_0 or at other points interior to the inner circle bounding the annulus. The $c_n's$ given by (2) are called the *Laurent coefficients of f about z_0*.

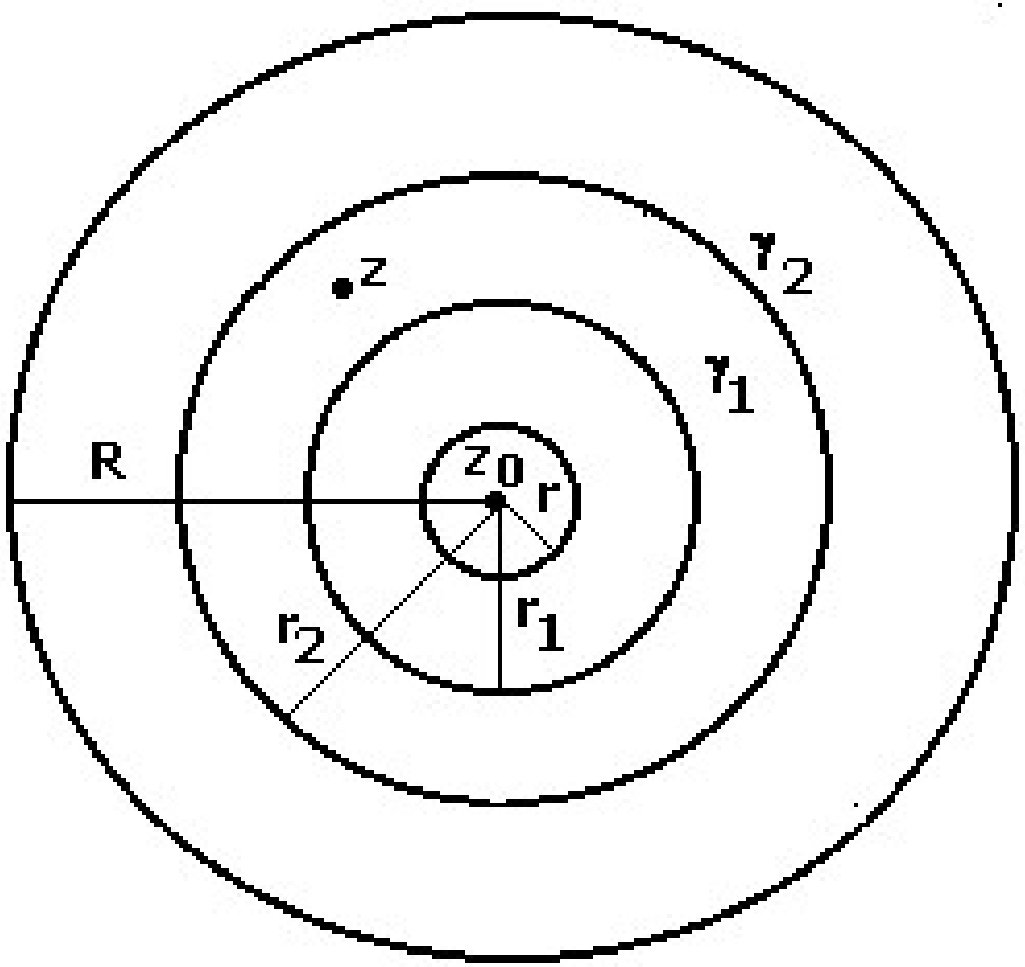
We may think of a Laurent series as a decomposition of $f(z)$ into a sum

$$f(z) = \sum_{n=-\infty}^{-1} c_n (z-z_0)^n + \sum_{n=0}^{\infty} c_n (z-z_0)^n = h(z) + g(z)$$

The part containing all the nonnegative powers of $z - z_0$ is a Taylor series about z_0 and defines a function g that is analytic on $|z - z_0| < R$. The part containing the negative powers of $z - z_0$ defines a function $h(z)$ that is not defined on $|z - z_0| < r$. This part contains any "bad features" of f within the inner circle.

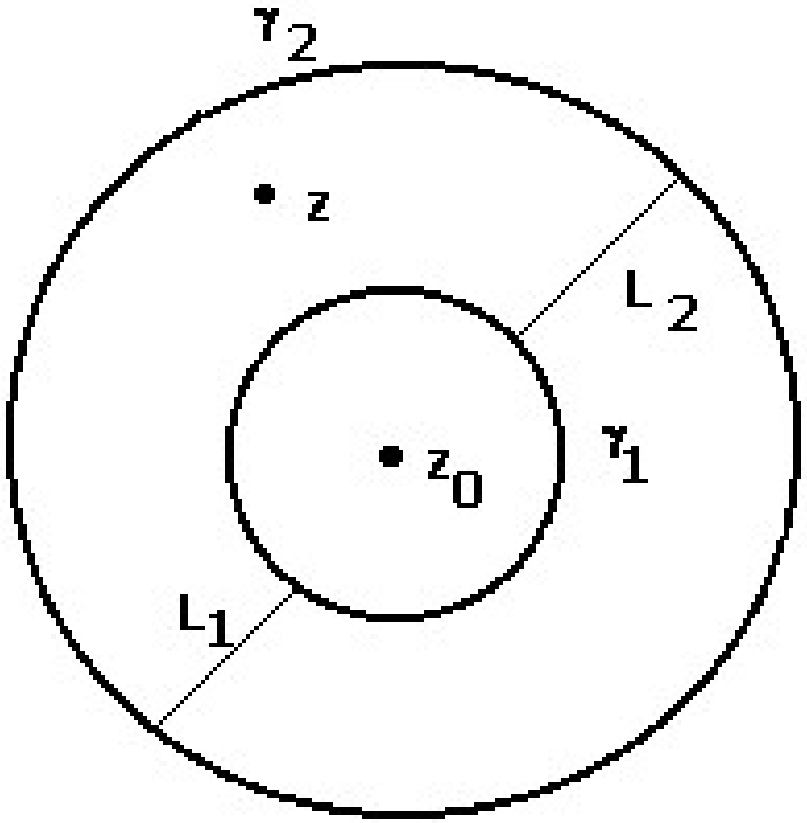
Proof of Laurent Expansion Theorem. Part 1

Let z be in the annulus. Choose r_1 and r_2 so that $r < r_1 < r_2 < R$ and so that the circle $\gamma_1 : |z - z_0| = r_1$ does not enclose z, while $\gamma_2 : |z - z_0| = r_2$ does enclose z.



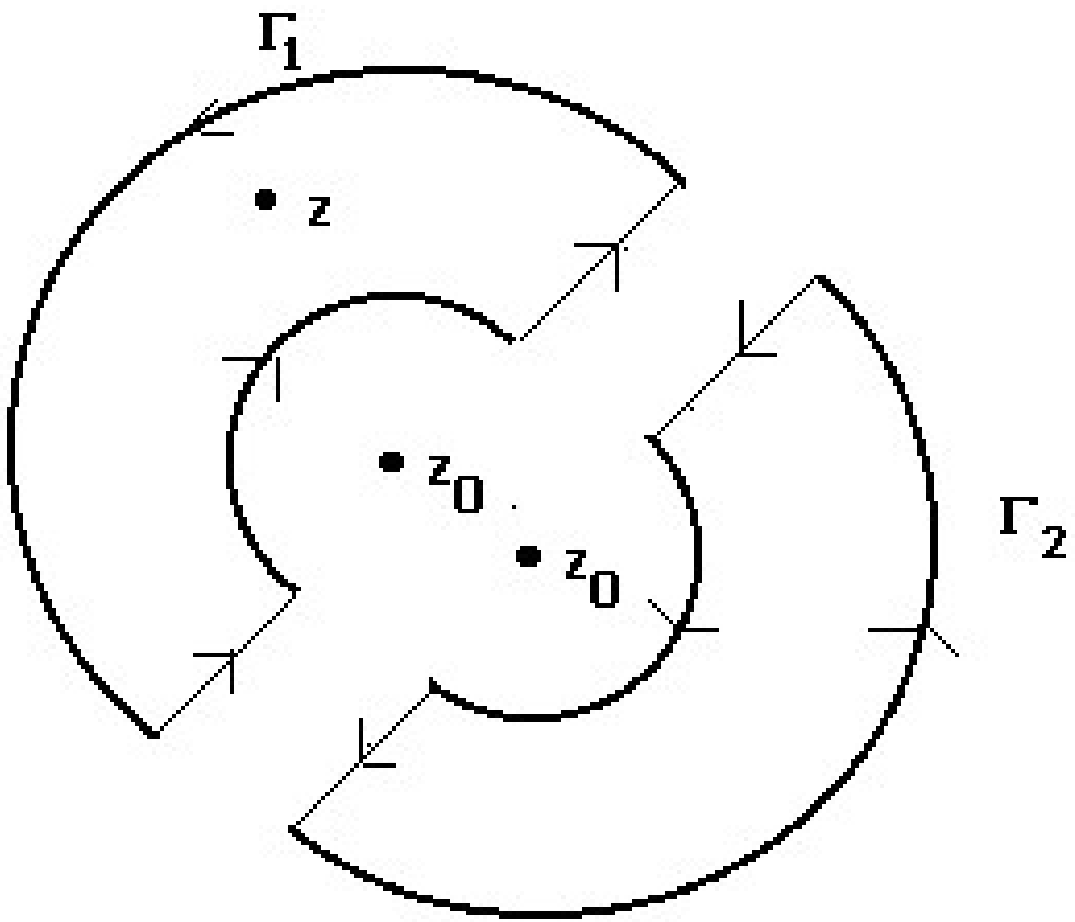
Proof of Laurent Expansion Theorem. Part 2

Insert the segments between γ_1 and γ_2 as shown below



Proof of Laurent Expansion Theorem. Part 3

forming two closed paths Γ_1 and Γ_2 as shown in the figure below.



Proof of Laurent Expansion Theorem. Part 4

Since f is differentiable on and in the interior of Γ_1, and this curve encloses z. Hence by the Cauchy Integral Formula we have that

$$f(z) = \frac{1}{2\pi i} \oint_{\Gamma_1} \frac{f(w)}{w-z} dw$$

By the Cauchy integral theorem

$$\oint_{\Gamma_2} \frac{f(w)}{w-z} dw = 0$$

because $f(z)/(w-z)$ is differentiable on and in the interior of Γ_2. Hence

$$f(z) = \frac{1}{2\pi i} \oint_{\Gamma_1} \frac{f(w)}{w-z} dw + \frac{1}{2\pi i} \oint_{\Gamma_2} \frac{f(w)}{w-z} dw$$

where both Γ_1 and Γ_2 are oriented counterclockwise.

Proof of Laurent Expansion Theorem. Part 5

Note that in the sum of the integrals we integrate over the two straight lines L_1 and L_2 in both directions, so that the net contribution is zero. However, with the contributions from L_1 and L_2 cancelling out we are thus integrating over γ_2 in the counterclockwise direction and γ_1 in the clockwise direction. Since $-\gamma_1$ is in the counterclockwise direction we may write the last expression as

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(w)}{w-z} dw - \frac{1}{2\pi i} \oint_{\gamma_1} \frac{f(w)}{w-z} dw \qquad (1)$$

Here the orientation on both circles is counterclockwise.

Proof of Laurent Expansion Theorem. Part 6

We may write the fraction in the first integral in (1) as

$$\frac{1}{w-z} = \frac{1}{w-z_0+z_0-z} = \frac{1}{w-z_0-(z-z_0)} = \frac{1}{w-z_0}\frac{1}{1-\left(\dfrac{z-z_0}{w-z_0}\right)}$$

$$= \frac{1}{w-z_0}\sum_{n=0}^{\infty}\left(\frac{z-z_0}{w-z_0}\right)^n = \sum_{n=0}^{\infty}\frac{1}{(w-z_0)^{n+1}}(z-z_0)^n$$

This last series is a convergent geometric series since w is on γ_2 and thus

$$\left|\frac{z-z_0}{w-z_0}\right| < 1.$$

Proof of Laurent Expansion Theorem. Part 7

For the fraction in the second integral in (1) we have

$$\frac{1}{w-z} = \frac{1}{w-z_0+z_0-z} = \frac{1}{w-z_0-(z-z_0)} = \frac{-1}{z-z_0}\frac{1}{1-\left(\dfrac{w-z_0}{z-z_0}\right)}$$

$$= \frac{-1}{z-z_0}\sum_{n=0}^{\infty}\left(\frac{w-z_0}{z-z_0}\right)^n = -\sum_{n=0}^{\infty}\frac{1}{(z-z_0)^{n+1}}(w-z_0)^n$$

This geometric expansion also converges since for w on γ_1 we have

$$\left|\frac{w-z_0}{z-z_0}\right| < 1.$$

Since f is differentiable on both curves and the convergence of both geometric series is absolute, we use (1) to write

Proof of Laurent Expansion Theorem. Part 8

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma_2} \left(\sum_{n=0}^{\infty} \frac{1}{(w - z_0)^{n+1}} f(w)\, dw \right) (z - z_0)^n$$

$$+ \frac{1}{2\pi i} \oint_{\gamma_1} \left(\sum_{n=0}^{\infty} (w - z_0)^n f(w)\, dw \right) \frac{1}{(z - z_0)^{n+1}}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n$$

$$+ \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{\gamma_1} (w - z_0)^n f(w)\, dw \right) \frac{1}{(z - z_0)^{n+1}}$$

Proof of Laurent Expansion Theorem. Part 9

In the last summation let $n = -m - 1$ to get

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n +$$

$$+ \sum_{m=-1}^{-\infty} \left(\frac{1}{2\pi i} \oint_{\gamma_2} \frac{f(w)}{(w - z_0)^{m+1}} dw \right) (z - z_0)^{-m}$$

Now use the deformation theorem to replace the integrals over γ_1 and γ_2 with integrals over γ so that we have

$$f(z) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

Thank you